Distributed local-displacement caps across architectures

Same prefix-q50 rule; each architecture calibrates its own affine modules from eight untreated updates.

MLP / CIFAR-10 | ResNet-32 / CIFAR-10 | Transformer / WikiText-2

Unchanged Adam
Distributed displacement cap (q50)

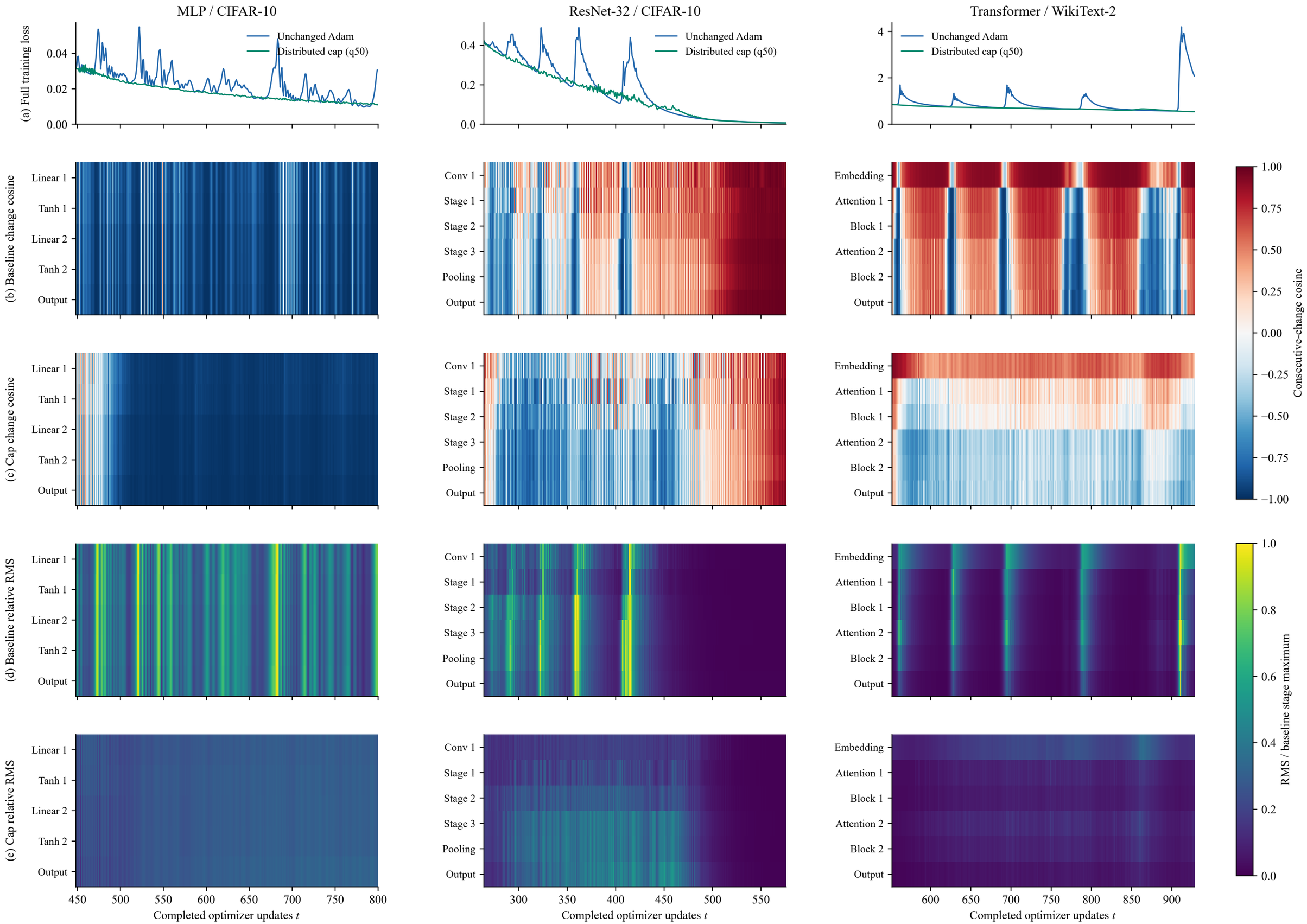
(a) Full training loss
(b) $\eta\hat{\kappa}_t$ (symlog)
(c) Raw-gradient cosine
(d) Full $\|g_t\|$ (log)
(e) Early change cosine
(f) Early RMS (log)
(g) Output-change cosine
(h) Output RMS (log)
(i) Update norm / proposal
(j) Mean fraction modified

Completed optimizer updates t

Full objectives; fixed diagnostic probes; outgoing representation changes; no smoothing. Each column uses its own unchanged baseline.
Early: MLP first tanh, ResNet conv1 output, Transformer first attention output. Loss and magnitude scales are independent by column.
The modified-fraction row averages targeted coordinate/neuron/block fractions; this is not a whole-model parameter fraction.
Undefined cosines and zero values on logarithmic axes appear as gaps.

Distributed caps: motion across captured representation stages

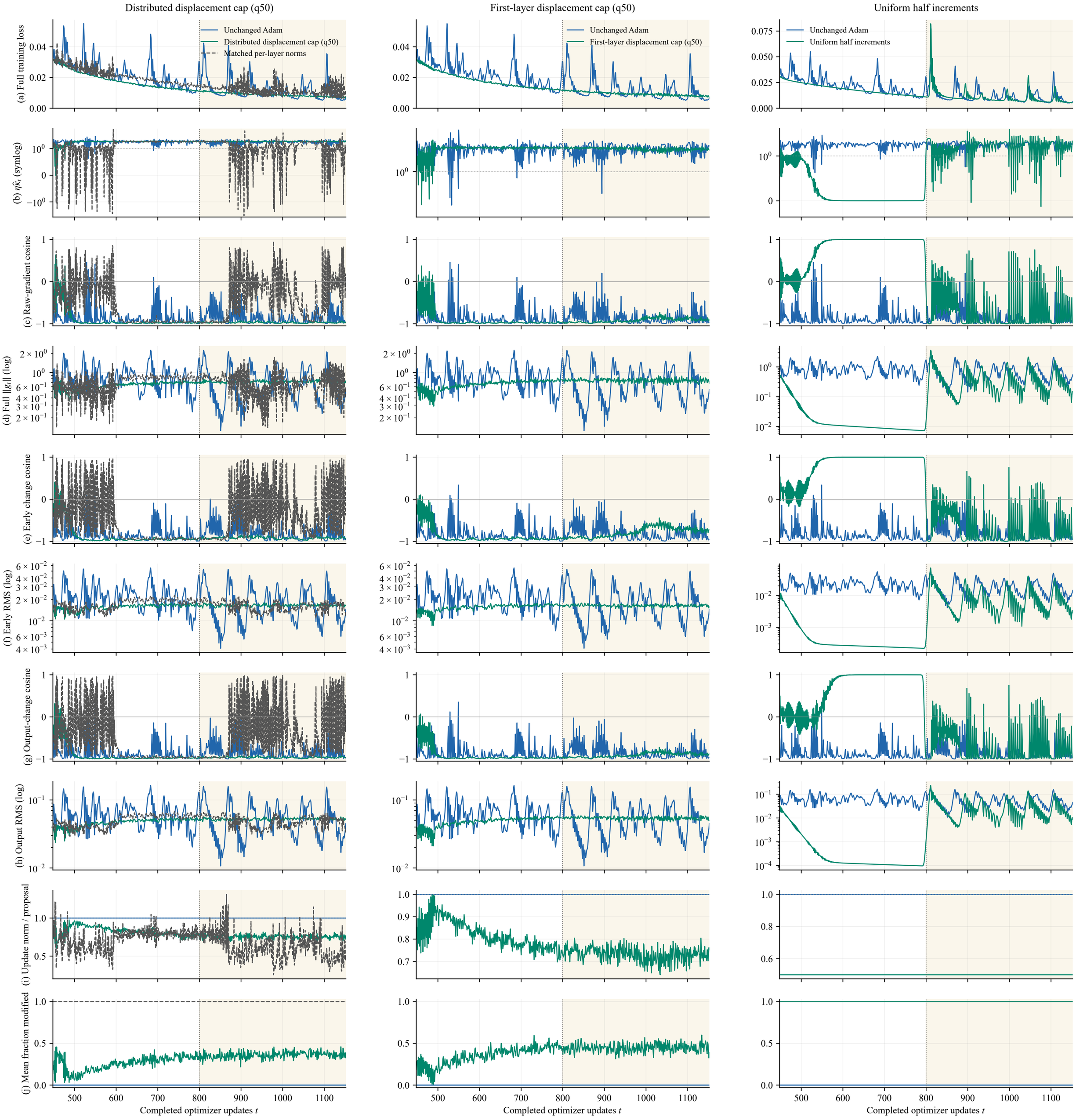
The same q50 local rule; compare reversal patterns and amplitudes with unchanged Adam.



Captured stages preserve all probe sample/token/spatial coordinates. No smoothing or new SVD decomposition.
Both RMS maps use each stage's baseline maximum in the displayed window; treatment amplitudes are not normalized away.
Same-worker unchanged Adam in every column. These maps show intervention effects on motion, not a unique source of instability.

MLP: later-window confirmation and matched norms

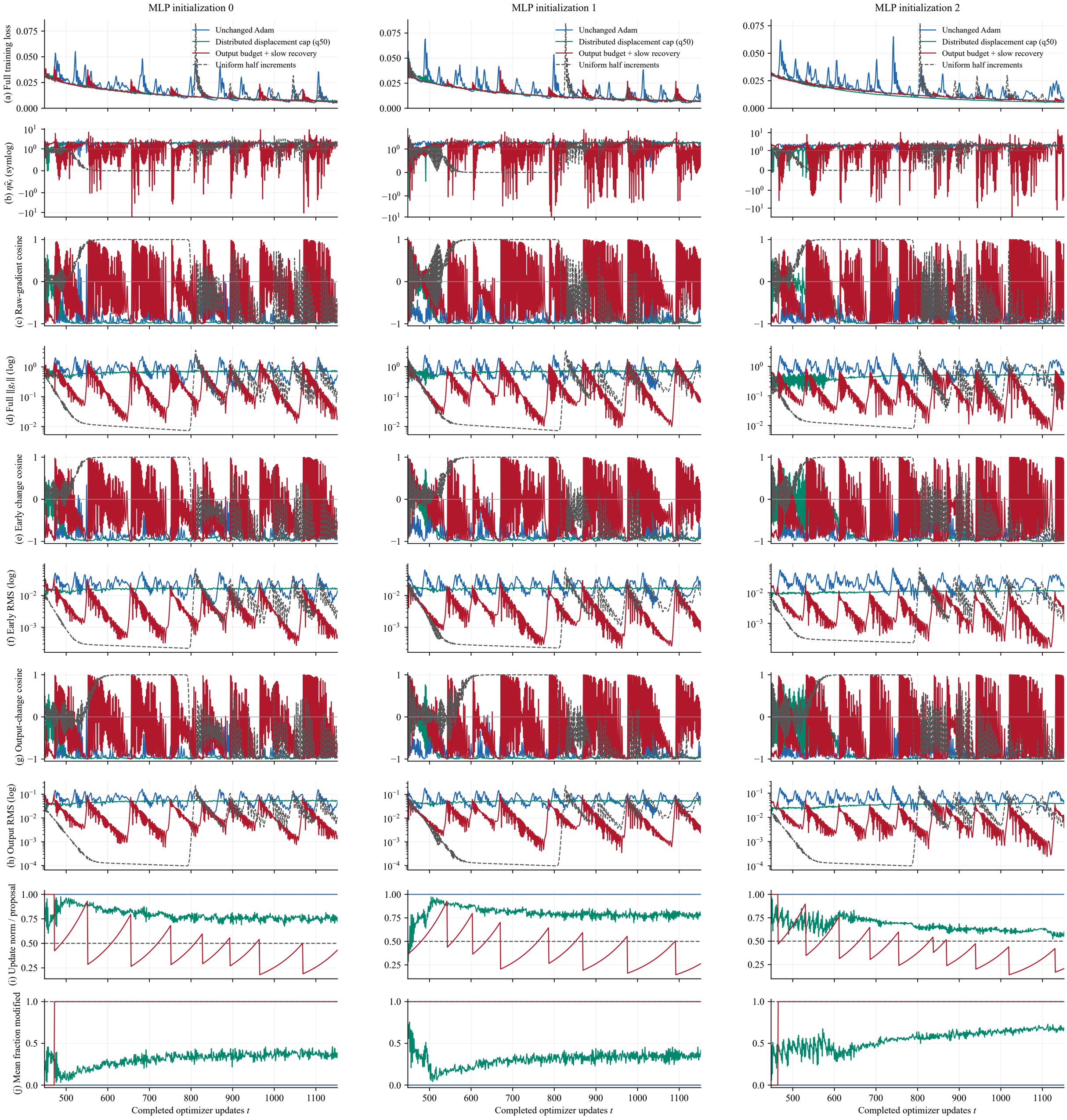
Continue the selected policies through 1152; unchanged Adam remains the baseline in every column.



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Shaded region 801-1152 was unseen at initial policy selection. The scalar-per-layer norm control is retrospective, not an online policy.
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MLP: replication across three initializations

Same rules and calibration length; initialization 0 selected the candidates, initializations 1/2 received no policy tuning.



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Slow recovery of an output-movement budget

Transfer the same q50 rule and 1% recovery limit; calibration thresholds differ by architecture.

MLP / CIFAR-10 | ResNet-32 / CIFAR-10 | Transformer / WikiText-2

Unchanged Adam
Output budget + slow recovery

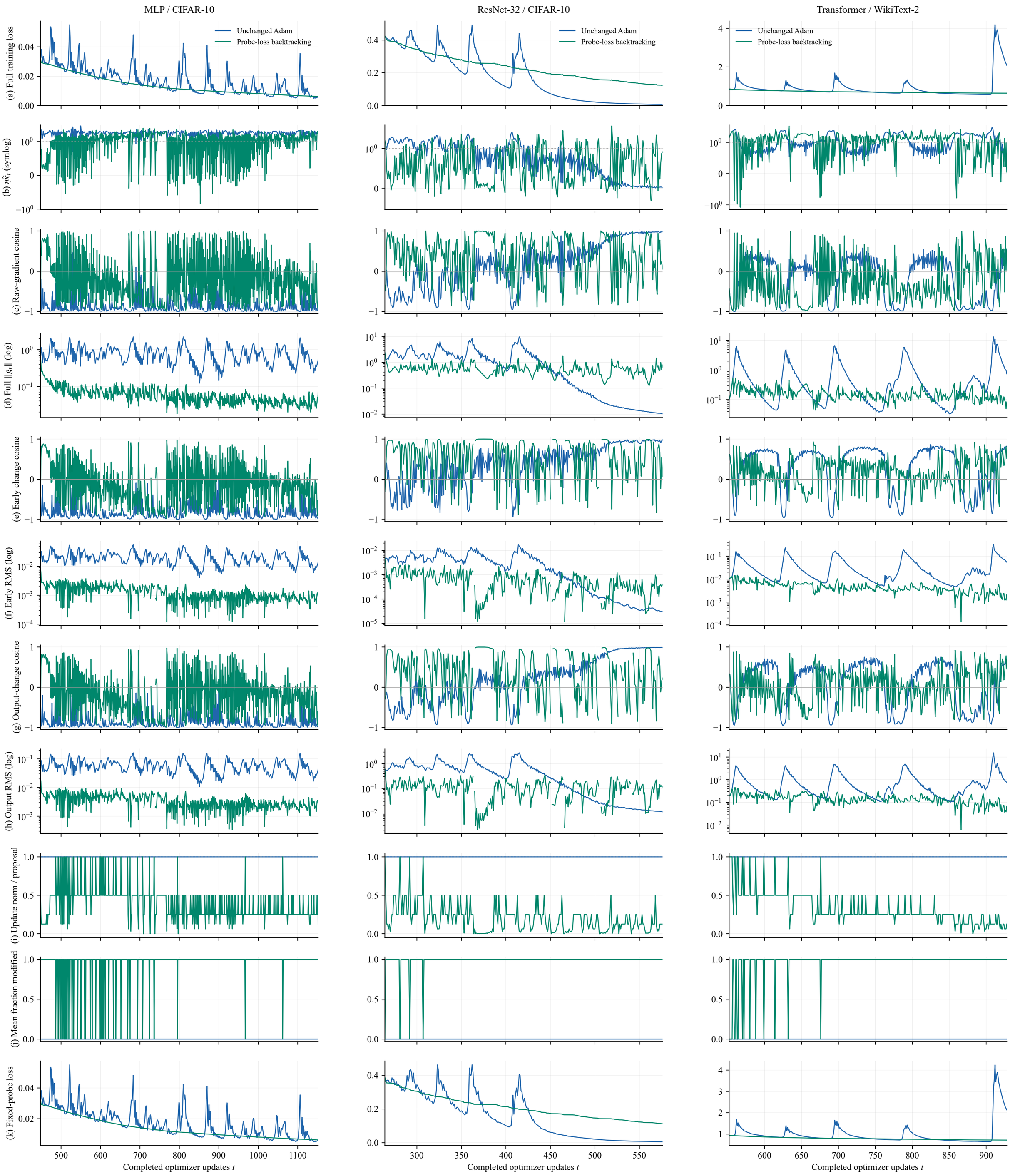
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A loss-aware comparator: probe-loss backtracking

Accept the first strictly decreasing dyadic candidate on the fixed probe; otherwise take a zero increment. Adam moments/clocks advance.



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Probe-loss acceptance covers all MLP examples, one ResNet normalization group, or 64 Transformer sequences; full-objective behavior is measured separately.
Undefined cosines and zero values on logarithmic axes appear as gaps.